

Exercise 2.3



Question 1 – Part (i)

If $z_1 = 1 - 3i$, $z_2 = -4i$, $z_3 = 5$, show that $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

LHS

$$\begin{aligned} z_1 + z_2 &= (1 - 3i) + (-4i) \\ &= 1 - 7i \end{aligned}$$

Now add z_3 :

$$\begin{aligned} (1 - 7i) + 5 \\ &= 6 - 7i \end{aligned}$$

$$\text{LHS} = 6 - 7i$$

RHS

$$\begin{aligned} z_2 + z_3 &= (-4i) + 5 \\ &= 5 - 4i \end{aligned}$$

Now add z_1 :

$$\begin{aligned} (1 - 3i) + (5 - 4i) \\ &= 6 - 7i \end{aligned}$$

$$\text{RHS} = 6 - 7i$$

Since LHS = RHS $\Rightarrow$ Hence Proved.

Question 1 – Part (ii)


If $z_1 = 1 - 3i$, $z_2 = -4i$, $z_3 = 5$, show that $(z_1 z_2) z_3 = z_1 (z_2 z_3)$

LHS

$$\begin{aligned} z_1 z_2 &= (1 - 3i)(-4i) \\ &= -4i + 12i^2 = -12 - 4i \end{aligned}$$

Multiply by $z_3 = 5$:

$$\begin{aligned} (-12 - 4i) \times 5 \\ &= -60 - 20i \end{aligned}$$

$$\text{LHS} = -60 - 20i$$

RHS

$$z_2 z_3 = (-4i)(5) = -20i$$

Now, $z_1(z_2 z_3)$

$$\begin{aligned} &= (1 - 3i)(-20i) \\ &= -20i + 60i^2 = -60 - 20i \end{aligned}$$

$$\text{RHS} = -60 - 20i$$

Since LHS = RHS $\Rightarrow$ Hence Proved.

Question 2 – Part (i)

If $z_1 = 3$, $z_2 = -7i$, $z_3 = 5 + 4i$, show that $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

LHS

$$\begin{aligned} z_2 + z_3 &= -7i + (5 + 4i) \\ &= 5 - 3i \end{aligned}$$

Now, $z_1(5 - 3i)$:

$$\begin{aligned} 3(5 - 3i) \\ &= 15 - 9i \end{aligned}$$

$$\text{LHS} = 15 - 9i$$

RHS

$$\begin{aligned} z_1z_2 &= 3(-7i) = -21i \\ z_1z_3 &= 3(5 + 4i) = 15 + 12i \end{aligned}$$

Now add:

$$\begin{aligned} -21i + 15 + 12i \\ &= 15 - 9i \end{aligned}$$

$$\text{RHS} = 15 - 9i$$

Since LHS = RHS $\Rightarrow$ Hence Proved.

Question 2 – Part (ii)


If $z_1 = 3$, $z_2 = -7i$, $z_3 = 5 + 4i$, show that $(z_1 + z_2)z_3 = z_1z_3 + z_2z_3$

LHS

$$z_1 + z_2 = 3 - 7i$$

Now, $(3 - 7i)(5 + 4i)$:

$$= 15 + 12i - 35i - 28i^2$$

$$= 15 - 23i + 28$$

$$= 43 - 23i$$

$$\text{LHS} = 43 - 23i$$

RHS

$$z_1z_3 = 3(5 + 4i) = 15 + 12i$$

$$z_2z_3 = (-7i)(5 + 4i)$$

$$= -35i - 28i^2 = -35i + 28$$

Now add:

$$15 + 12i + 28 - 35i = 43 - 23i$$

$$\text{RHS} = 43 - 23i$$

Since LHS = RHS $\Rightarrow$ Hence Proved.

Question 3 — Additive & Multiplicative Inverse

Given $z_1 = 2 + 5i$, $z_2 = -3 - 4i$, $z_3 = 1 + i$. Formula: $-z$ and $1/z = (a - bi)/(a^2 + b^2)$

(i) $z_1 = 2 + 5i$

Additive Inverse ($-z$):

$$-z_1 = -(2 + 5i)$$

$$\mathbf{-2 - 5i}$$

Multiplicative Inverse ($1/z$):

$$\begin{aligned} 1/(2 + 5i) &= (2 - 5i)/(2^2 + 5^2) \\ &= (2 - 5i)/29 \end{aligned}$$

$$\mathbf{2/29 - 5/29 i}$$

Question 3 – Additive & Multiplicative Inverse

Given $z_1 = 2 + 5i$, $z_2 = -3 - 4i$, $z_3 = 1 + i$. Formula: $-z$ and $1/z = (a - bi)/(a^2 + b^2)$

(ii) $z_2 = -3 - 4i$

Additive Inverse ($-z$):

$$-z_2 = -(-3 - 4i)$$

$$3 + 4i$$

Multiplicative Inverse ($1/z$):

$$\begin{aligned} 1/(-3-4i) &= (-3+4i)/((-3)^2+(-4)^2) \\ &= (-3 + 4i)/25 \end{aligned}$$

$$-3/25 + 4/25 i$$

Question 3 – Additive & Multiplicative Inverse

Given $z_1 = 2 + 5i$, $z_2 = -3 - 4i$, $z_3 = 1 + i$. Formula: $-z$ and $1/z = (a - bi)/(a^2 + b^2)$

(iii) $z_3 = 1 + i$

Additive Inverse ($-z$):

$$-z_3 = -(1 + i)$$

$$\mathbf{-1 - i}$$

Multiplicative Inverse ($1/z$):

$$1/(1 + i) = (1 - i)/(1^2 + 1^2)$$

$$= (1 - i)/2$$

$$\mathbf{1/2 - 1/2 i}$$